

Use Case Pay Order(VietQR)

1. Use case code

UC232

2. Brief Description

This use case describes the interaction between the customer, the AIMS software, and the VietQR system when the customer makes a payment for an order using a QR code. The process begins when the system displays invoice information along with the payment method and ends when the transaction is successfully recorded.

3. Actors

3.1 Customer

3.2 VietQR

4. Preconditions

- The Customer has completed entering shipping information and confirming the order (finishing the preparation steps).
- The system has calculated the shipping fee and generated temporary invoice.

5. Basic Flow of Events

1. Customer request to pay order
2. System display and save a temporary invoice information
3. System call VietQR's Get Token API to obtain an Access Token to access the system
4. VietQR return access token to System
5. System call Generate VietQR Code API use token to create QR code for payment base on total money of invoice
6. VietQR return QR code can be image or data string to System.
7. System show the QR code and detail invoice.
8. Customer scan QR code
9. Customer confirm the payment
10. VietQR record successful transactions send notification to System.
11. System receive callback data and recheck order ID and amount
12. System update status order to pending and store detail transaction
13. System display successful payment with order information and transaction code
14. System clear cart
15. System send email confirm invoice and transaction information to Customer's email

6. Alternative flows

Table 1 - Alternative flows of events for UC “Pay Order By Credit Card”

No	Location	Condition	Action	Resume location
1.	At step 7	If customer want to change payment method	System change to usecase “Pay Order by Credit Card”.	Use case end
2.	At step 8	If the customer want to change order information.	System call “Place Order” use case	Use case end
3.	At step 9	Transaction is failed	System notify the failure and allow customer try scanning again	At step 7
4.	At step 10	If Network timeout / No callback received	Customer clicks "I have paid". System calls "Check Transaction" API to verify.	At step 11
5.	At Step 11	If QR code expired	System notify expired QR code, request to refresh	At step 3

7. Input data**Table 2 - Input data of UC “Pay order by credit card”**

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Access Token	Token for API authentication	Yes	Validated by VietQR	eyJhbGciOi..
2.	Amount	Total money to pay	Yes	Positive integer	1,200,000
3.	Order ID	Unique ID for reconciliation	Yes	Matches existing order	ORDER_123
4.	Callback Data	Notification from VietQR	Yes	Valid signature/checksum	JSON Object

8. Output data

Table 3 - Output data of UC “Pay order by credit card”

No	Data fields	Description	Display format	Example
1.	QR Code	Visual code for scanning	Image/String	
2.	Total amount	Final amount paid	Vietnamese currency (VNĐ)	1.200.000 VNĐ
3.	Transaction ID	Provided by VietQR	String	VQR123456
4.	Transaction date	Date of successful payment	dd/mm/yyyy	16/03/2026

9. Postconditions

- A new order is created with a "Pending" status.
- System sent the invoice and payment transaction information to the customer's email.
- System save recorded of payment transaction information and order detail in the history.
- System clear Customer's cart.